



Design and Analysis of Algorithms

Vandana M L

Department of Computer Science & Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

Fundamentals of Algorithmic Problem Solving

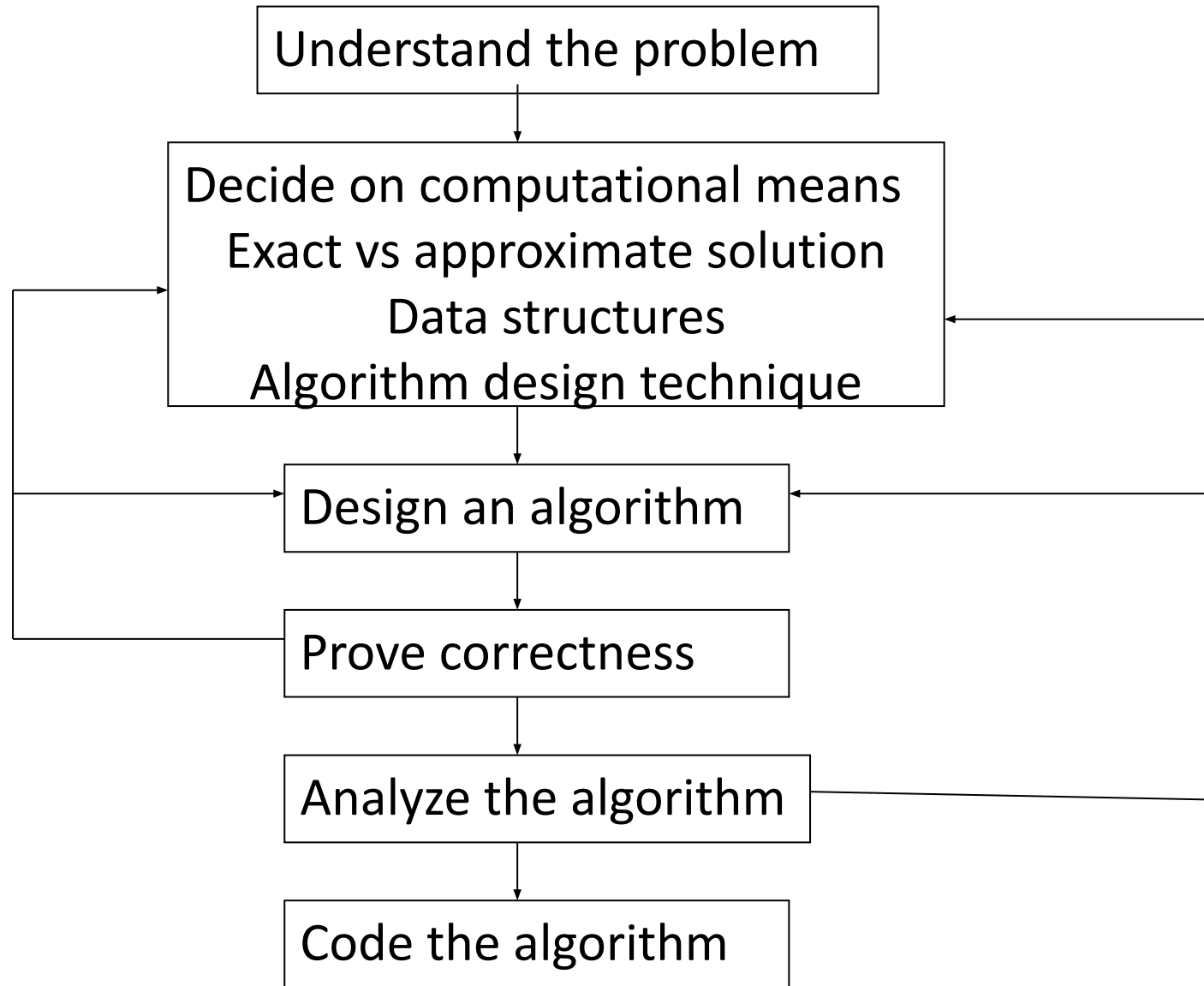
Slides courtesy of **Anany Levitin**

Vandana M L

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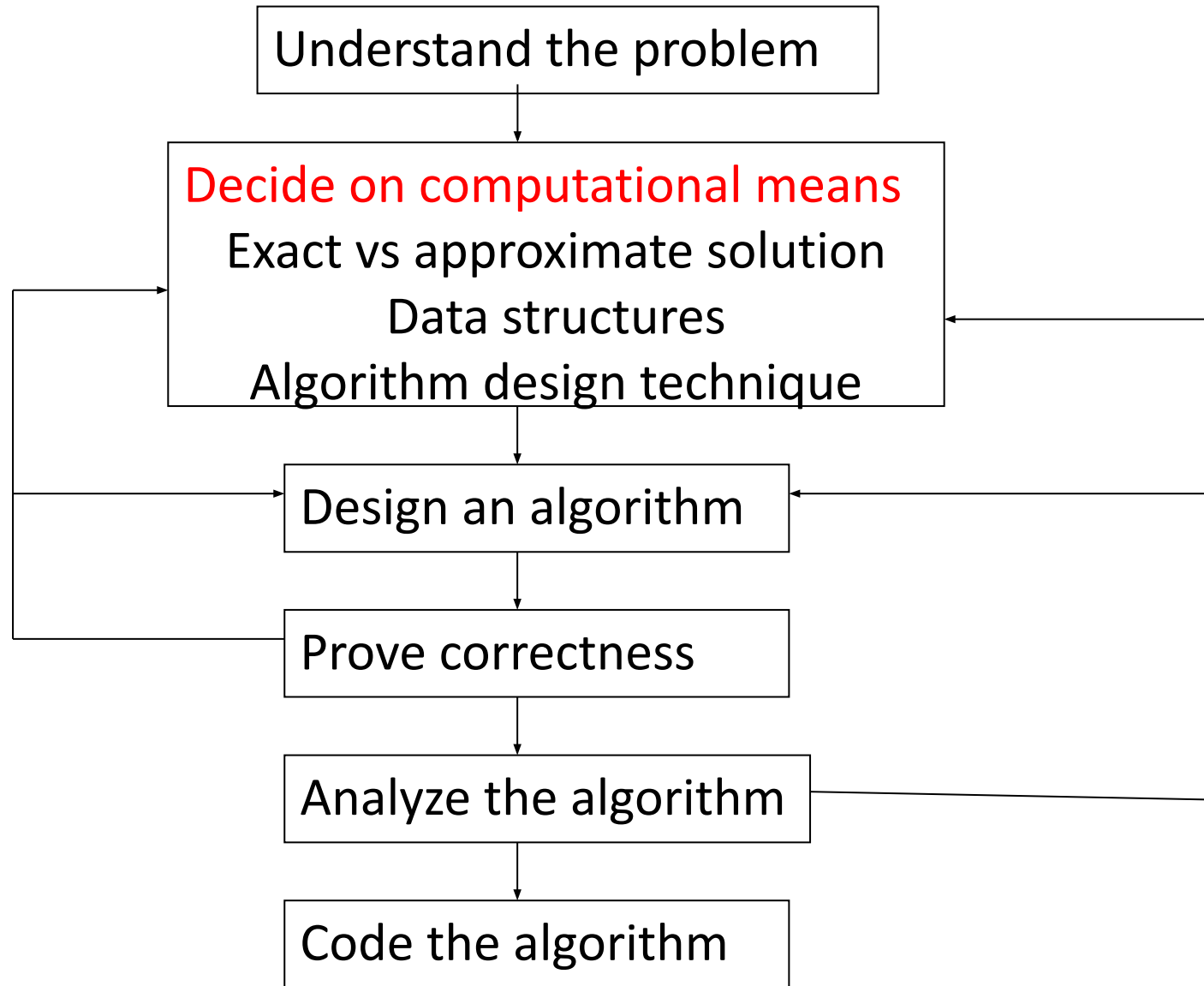
Design and Analysis of Algorithms

Algorithm Design and Analysis Process



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Computational Means



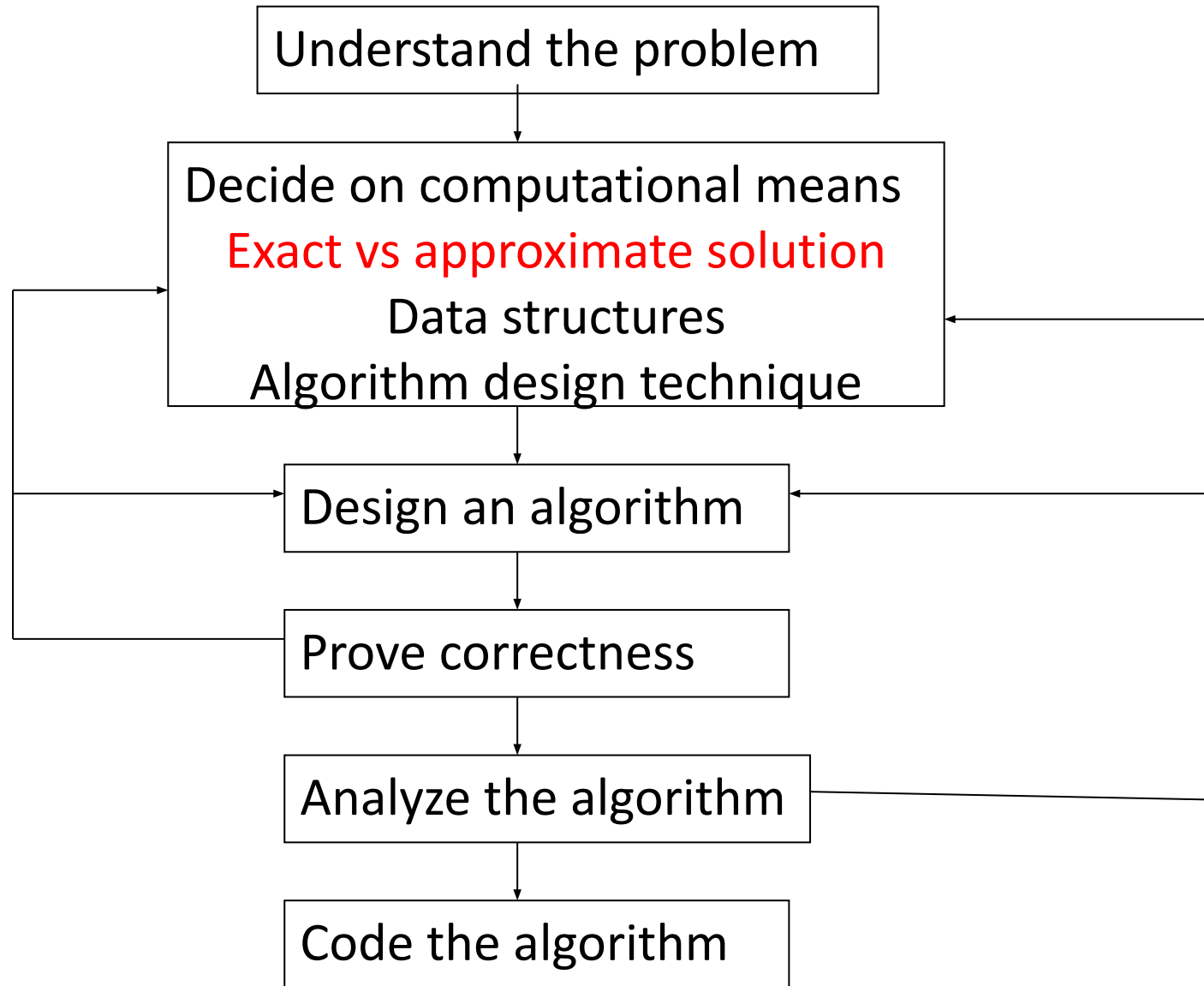
Computational Device the algorithm is intended for

RAM Sequential Algorithms

PRAM Parallel Algorithms

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Algorithm Design and Analysis Process



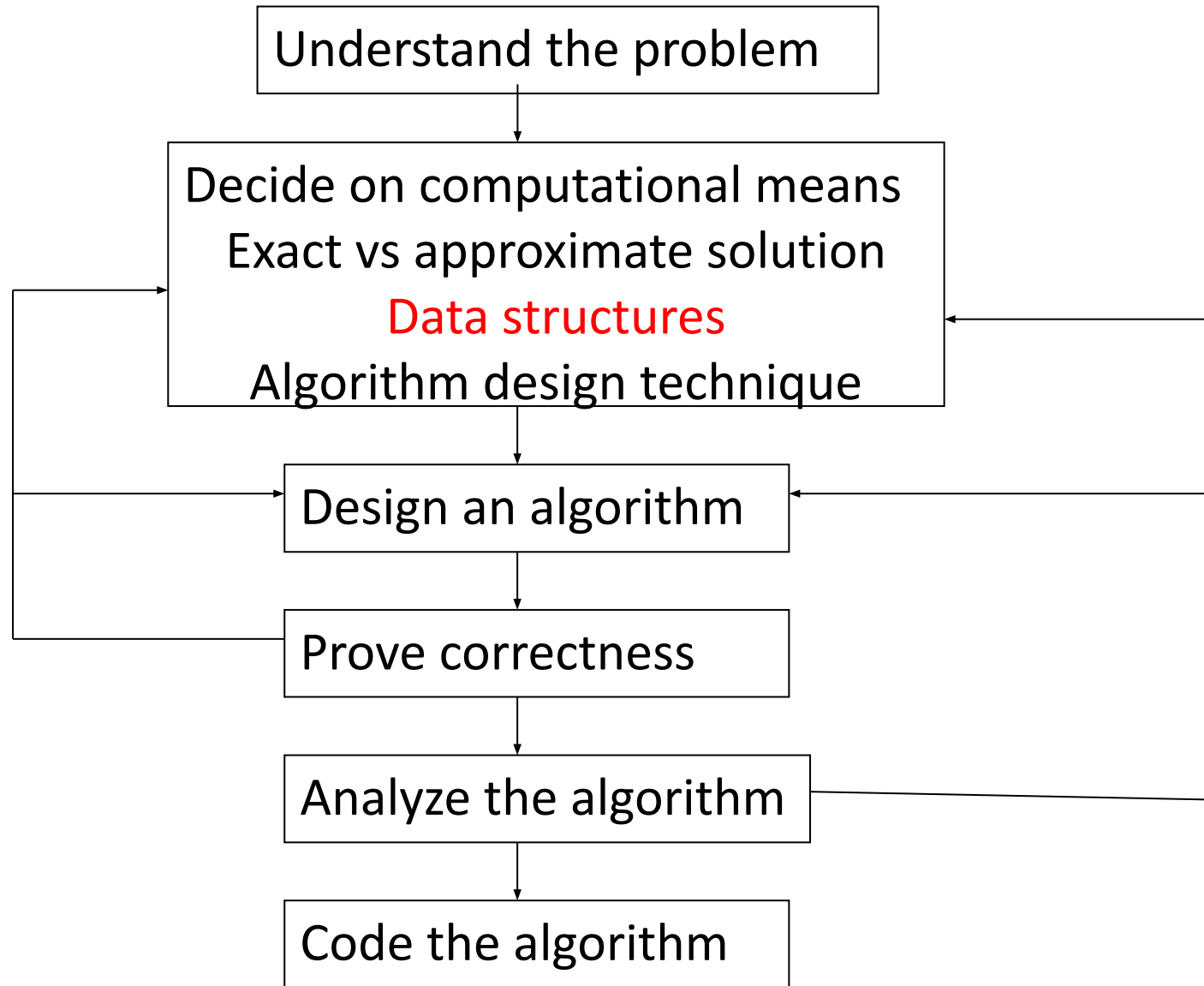
Travelling Salesman Problem

NP complete!!!

Approximate algorithm can be used to solve it

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Algorithm Design and Analysis Process



- Linear

 - Linear list, Stack, Queues

- Non Linear

 - Trees, Graphs

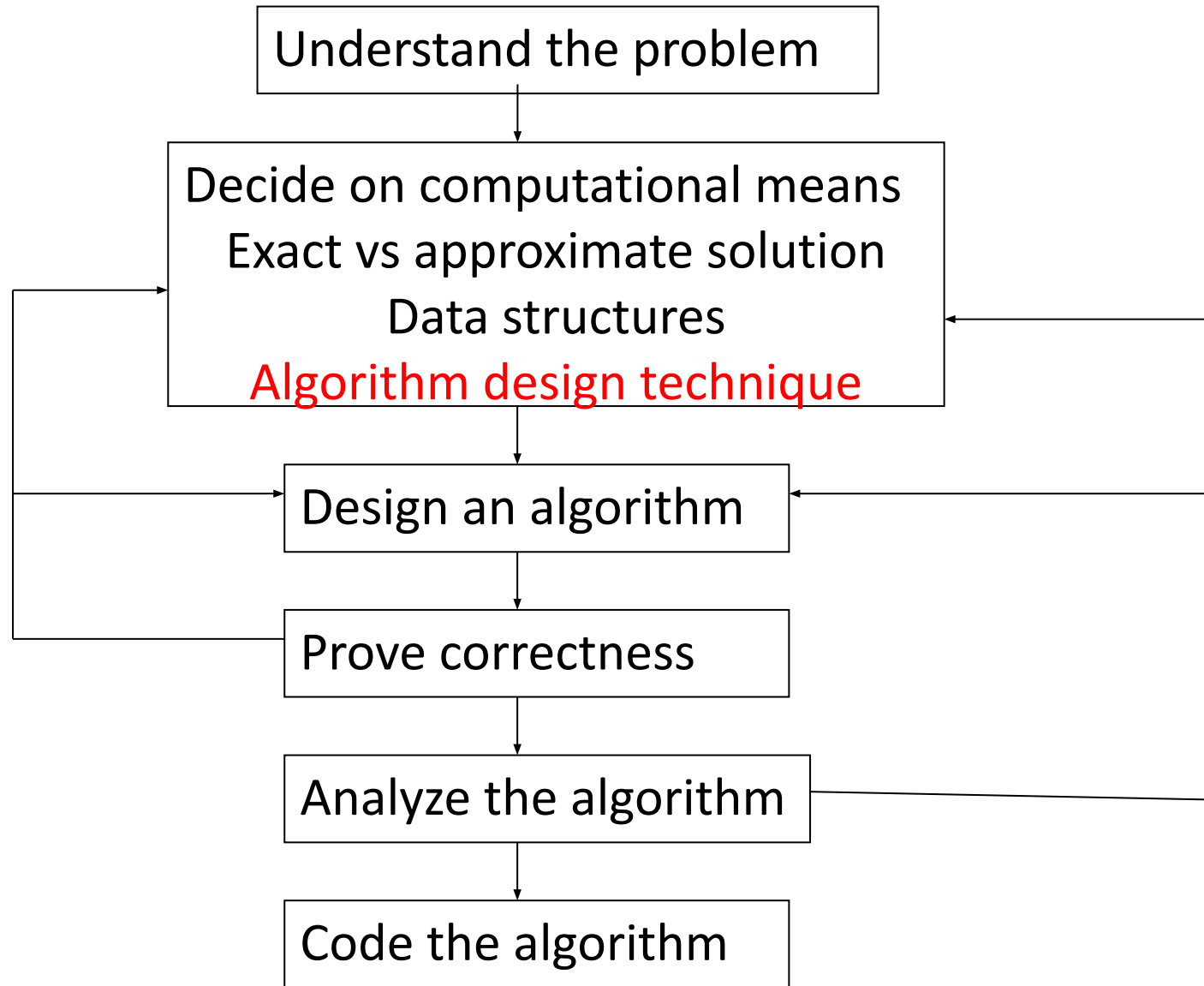
Choice of Data structure for solving a problem using an algorithm may dramatically impact its time complexity

Dijkstra Algorithm

 - $O(V \log V + E)$ with Fibonacci heap

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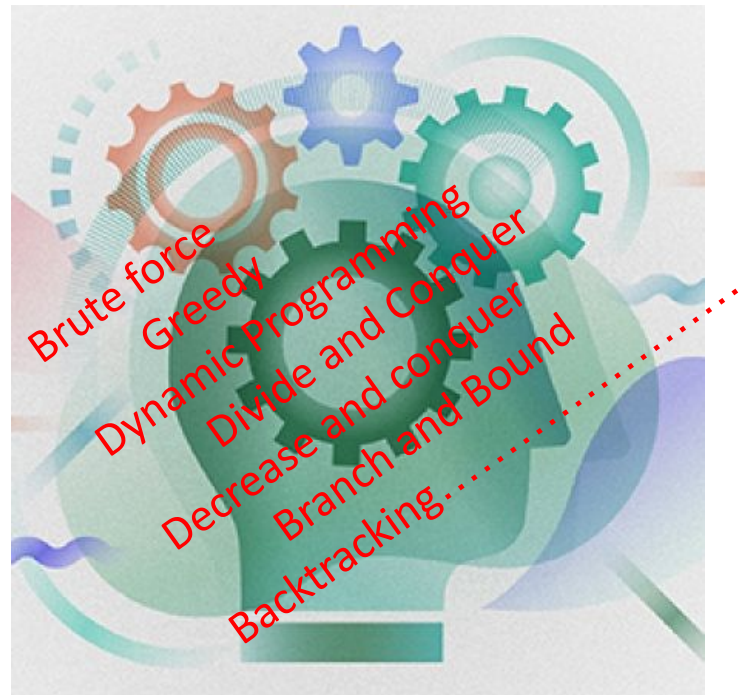
Algorithm Design and Analysis Process



General approach to solving problems algorithmically that is applicable to variety of problems from different areas of computing

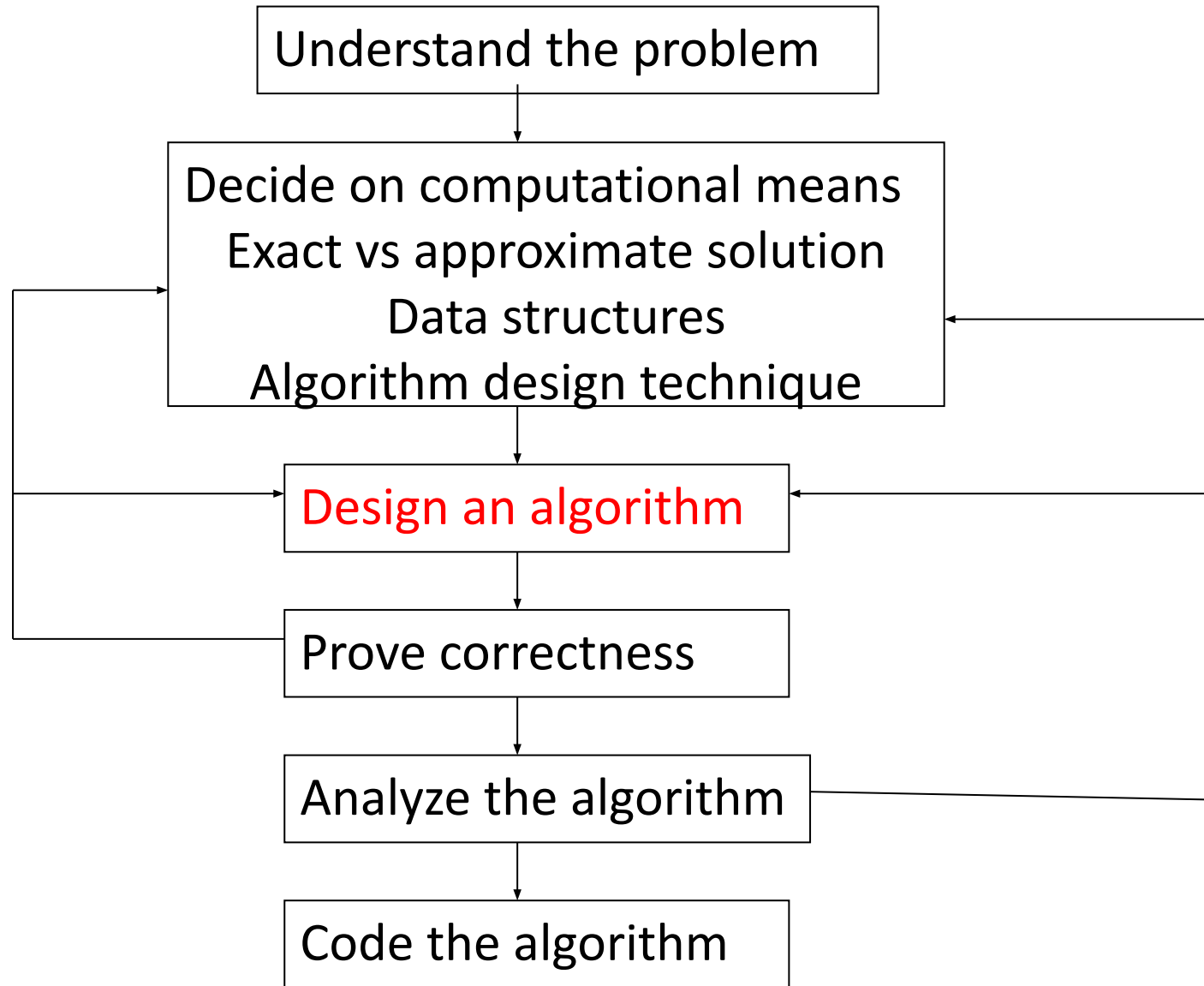
ADT serves as heuristic for designing algorithms for new problems for which no satisfactory algorithm exists!!!

Algorithm Designer's Toolkit



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Algorithm Design and Analysis Process



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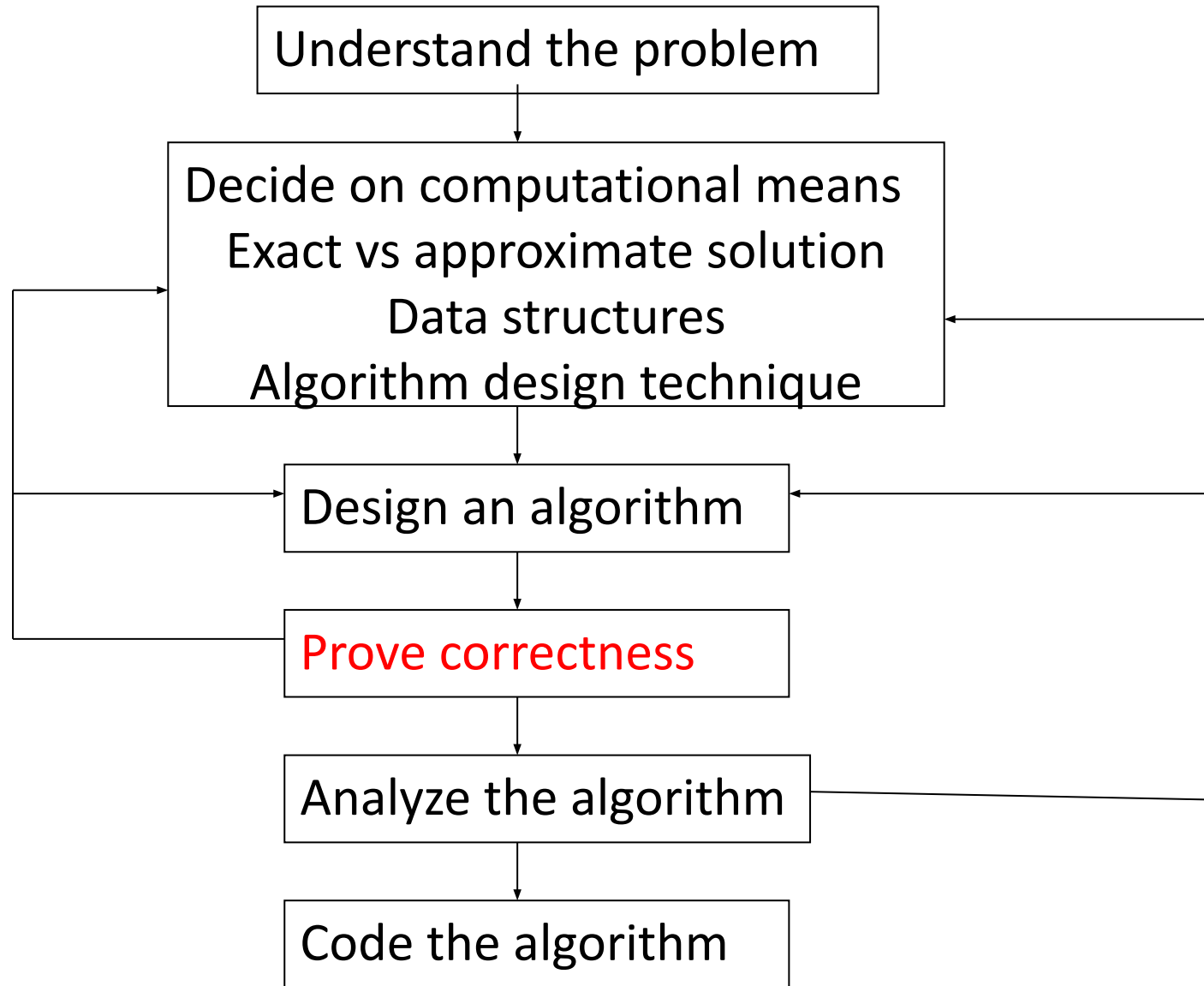
Specifying an algorithm



- Natural Language
- Pseudo Code
- Flowchart

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Exact algorithms

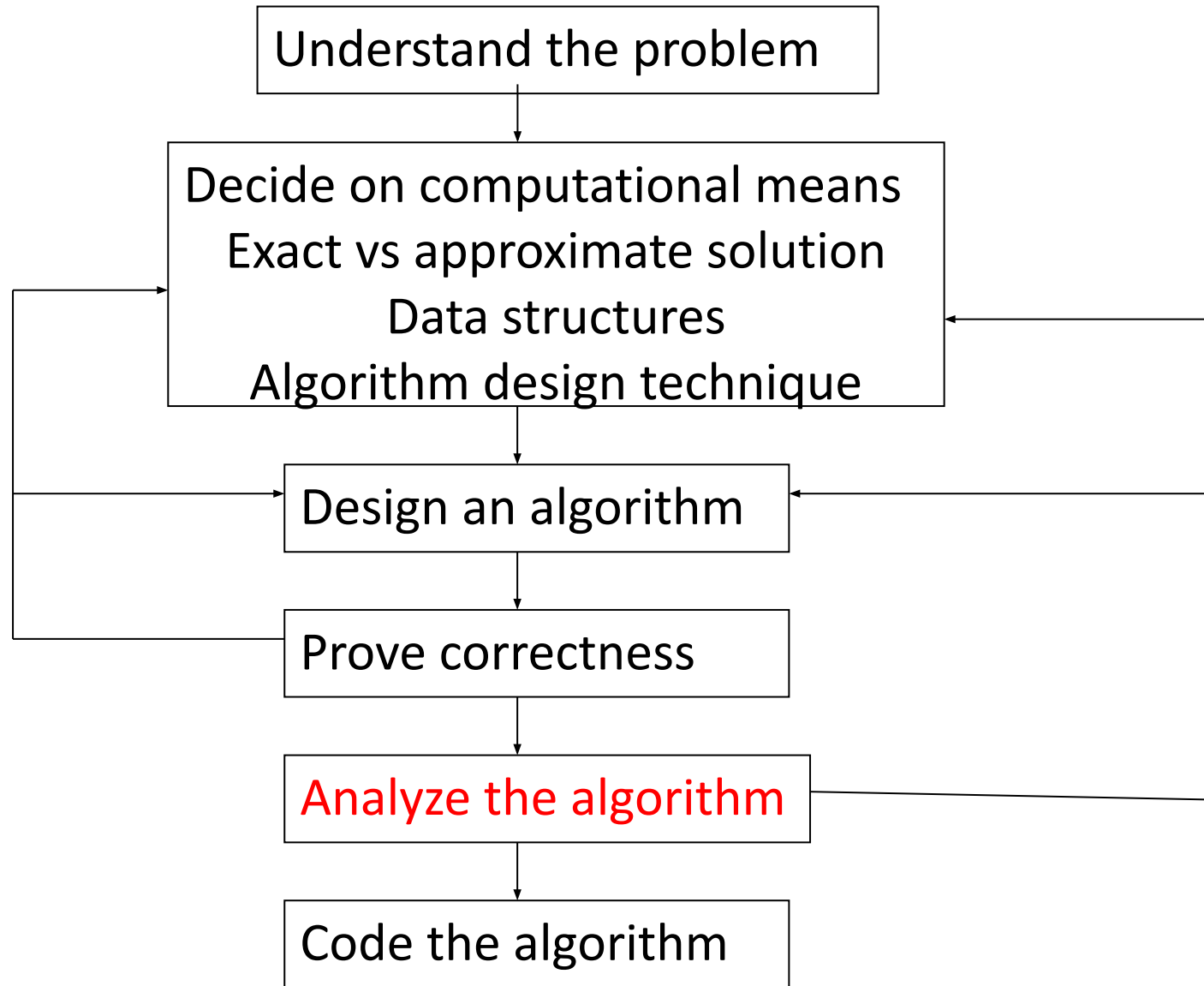
Proving that algorithm yields a correct result for legitimate input in finite amount of time

Approximation algorithms

Error produced by algorithm does not exceed a predefined limit

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Algorithm Design and Analysis Process



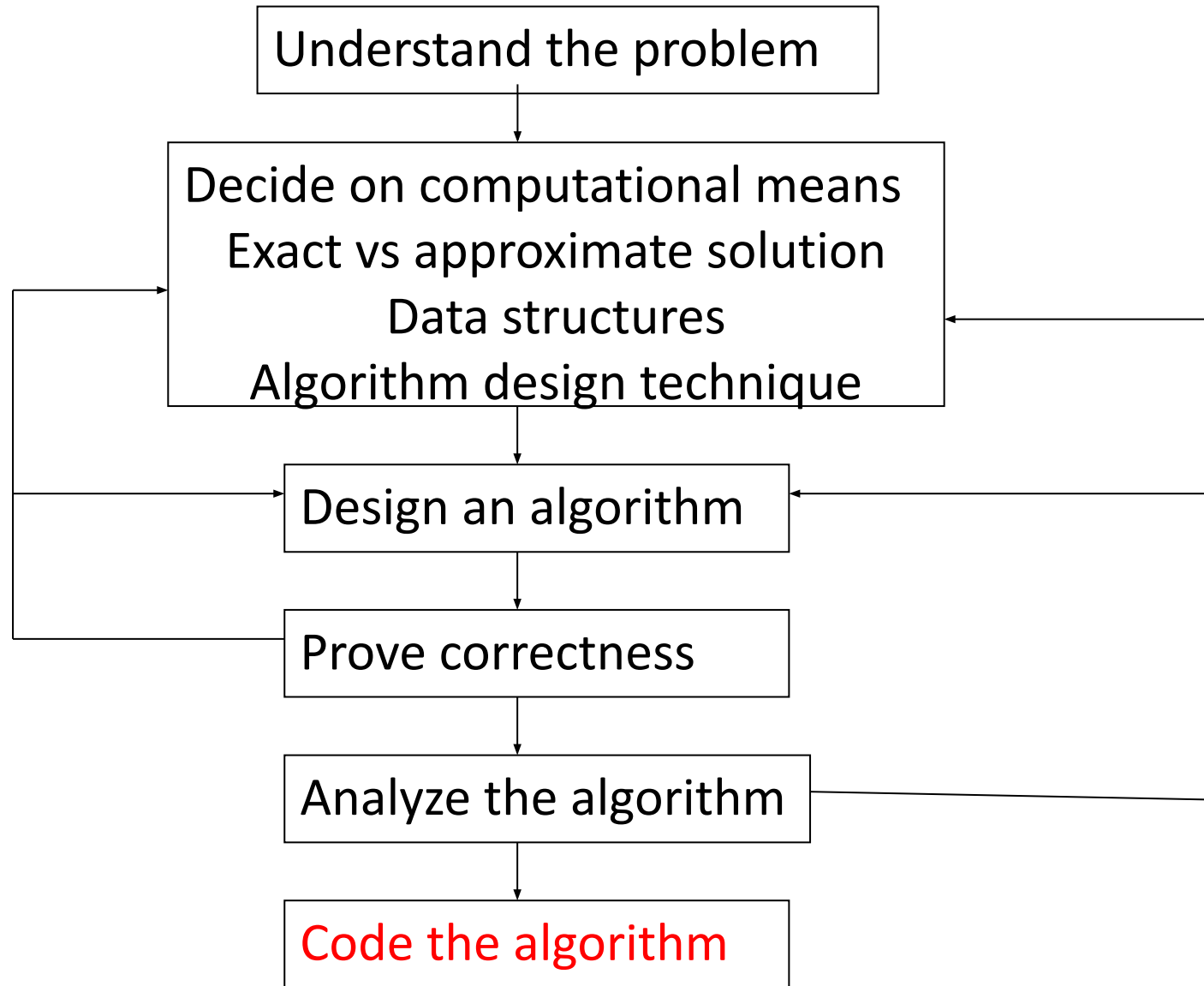
- Efficiency
 - Time efficiency
 - Space efficiency

- Simplicity

- Generality
 - Design an algorithm for the problem posed in more general terms
 - Design an algorithm that can handle a range of inputs that is natural for the problem at hand

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Algorithm Design and Analysis Process



- Efficient implementation
- Correctness of program
 - **Mathematical Approach**: Formal verification for small programs
 - **Practical Methods**: Testing and Debugging
- Code optimization

Design and Analysis of Algorithms

Multiple Choice Questions



- Efficient implementation
- Correctness of program
 - **Mathematical Approach**: Formal verification for small programs
 - **Practical Methods**: Testing and Debugging
- Code optimization

In the algorithm design and analysis process, why is deciding on computational means performed *before* designing the algorithm?

- a) To simplify coding
- b) To choose between exact and approximate solutions, data structures, and design paradigms
- c) To reduce the number of correctness proofs
- d) To estimate memory usage

Answer

- b) To choose between exact and approximate solutions, data structures, and design paradigms

Which scenario most strongly justifies choosing an approximate solution over an exact solution?

- a) When the input size is small
- b) When the problem has a polynomial-time exact solution
- c) When the problem is NP-complete and exact solutions are computationally infeasible
- d) When memory usage must be minimized

Answer

- c) When the problem is NP-complete and exact solutions are computationally infeasible

Why is the Travelling Salesman Problem (TSP) typically solved using approximation algorithms in practice?

- a) Exact algorithms do not exist
- b) Exact algorithms violate correctness
- c) TSP has exponential-time exact solutions and is NP-complete
- d) Approximate solutions always produce optimal tours

Answer

- c) TSP has exponential-time exact solutions and is NP-complete

Which of the following best explains the impact of data structure choice on algorithm performance?

- a) Data structures only affect space complexity
- b) Data structures affect only implementation, not design
- c) An inappropriate data structure can change polynomial-time algorithms into exponential-time ones
- d) Data structures can significantly affect time complexity without changing the algorithmic logic

Answer

- d) Data structures can significantly affect time complexity without changing the algorithmic logic

Which step in the algorithm design process ensures that the algorithm always produces correct output for all valid inputs?

- a) Decide on computational means
- b) Analyze the algorithm
- c) Prove correctness
- d) Code the algorithm

Answer

- c) Prove correctness

Why does the algorithm design process include feedback loops between steps such as design, correctness proof, and analysis?

- a) To reduce the number of algorithms designed
- b) To allow revisions when correctness or efficiency requirements are not met
- c) To optimize coding style
- d) To ensure hardware compatibility

Answer

- b) To allow revisions when correctness or efficiency requirements are not met

Which of the following situations most strongly suggests the use of a parallel (PRAM) model instead of a RAM model?

- a) Algorithms with recursive structure
- b) Problems where input size is small
- c) Problems that can exploit simultaneous operations to reduce time complexity
- d) Problems requiring exact solutions

Answer

- c) Problems that can exploit simultaneous operations to reduce time complexity



THANK YOU

Vandana M L

Department of Computer Science & Engineering

vandanamd@pes.edu